

A Technical Introduction to Autoencoder

Section 1

If linear activations are used, or only a single sigmoid hidden layer, then the optimal solution to an autoencoder is strongly related to principal component analysis (PCA). The weights of an autoencoder with a single hidden layer of size p (where p is less than the size of the input) span the same vector subspace as the one spanned by the first p principal components, and the output of the autoencoder is an orthogonal projection onto this subspace. The autoencoder weights are not equal to the principal components, and are generally not orthogonal, yet the principal components may be recovered from them using the singular value decomposition. However, the potential of autoencoders resides in their non-linearity, allowing the model to learn more powerful generalizations compared to PCA, and to reconstruct the input with significantly lower information loss.

Section 2

Definition An autoencoder is defined by the following components: Two sets: the space of encoded messages Z ; the space of decoded messages X . Typically X and Z are Euclidean spaces, that is, $X = \mathbb{R}^m$, $Z = \mathbb{R}^n$ with $m > n$. Two parametrized families of functions: the encoder family $E_\phi : X \rightarrow Z$, parametrized by ϕ ; the decoder family $D_\theta : Z \rightarrow X$, parametrized by θ . For any $x \in X$, we usually write $z = E_\phi(x)$, and refer to it as the code, the latent variable, latent representation, latent vector, etc. Conversely, for any $z \in Z$, we usually write $x' = D_\theta(z)$, and refer to it as the (decoded) message. Usually, both the encoder and the decoder are defined as multilayer perceptrons (MLPs). For example, a one-layer-MLP encoder E_ϕ is:

Section 3

Variational autoencoders (VAEs) belong to the families of variational Bayesian methods. Despite the architectural similarities with basic autoencoders, VAEs are architected with different goals and have a different mathematical formulation. The latent space is, in this case, composed of a mixture of distributions instead of fixed vectors. Given an

input dataset x characterized by an unknown probability function $P(x)$ and a multivariate latent encoding vector z , the objective is to model the data as a distribution $p_\theta(x)$, with θ defined as the set of the network parameters so that $p_\theta(x) = \int z p_\theta(x, z) dz$.

Section 4

Training Geoffrey Hinton developed the deep belief network technique for training many-layered deep autoencoders. His method involves treating each neighboring set of two layers as a restricted Boltzmann machine so that pretraining approximates a good solution, then using backpropagation to fine-tune the results. Researchers have debated whether joint training (i.e. training the whole architecture together with a single global reconstruction objective to optimize) would be better for deep auto-encoders. A 2015 study showed that joint training learns better data models along with more representative features for classification as compared to the layerwise method. However, their experiments showed that the success of joint training depends heavily on the regularization strategies adopted.

Section 5

History (Oja, 1982) noted that PCA is equivalent to a neural network with one hidden layer with identity activation function. In the language of autoencoding, the input-to-hidden module is the encoder, and the hidden-to-output module is the decoder. Subsequently, in (Baldi and Hornik, 1989) and (Kramer, 1991) generalized PCA to autoencoders, a technique which they termed "nonlinear PCA". Immediately after the resurgence of neural networks in the 1980s, it was suggested in 1986 that a neural network be put in "auto-association mode". This was then implemented in (Harrison, 1987) and (Elman, Zipser, 1988) for speech and in (Cottrell, Munro, Zipser, 1987) for images. In (Hinton, Salakhutdinov, 2006), deep belief networks were developed. These train a pair restricted Boltzmann machines as encoder-decoder pairs, then train another pair on the latent representation of the first pair, and so on. The first applications of AE date to early 1990s. Their most traditional application was dimensionality reduction or feature learning, but the concept became widely used for learning generative models of data. Some of the most powerful AIs in the 2010s involved autoencoder modules as a component of larger AI systems, such as VAE in Stable Diffusion, discrete VAE in Transformer-based image generators like DALL-E 1, etc. During the early days, when the terminology was uncertain, the autoencoder has also been called identity mapping, auto-associating, self-supervised backpropagation, or Diabolo network.

Section 6

Information retrieval Information retrieval benefits particularly from dimensionality reduction in that search can become more efficient in certain kinds of low dimensional spaces. Autoencoders were indeed applied to semantic hashing, proposed by Salakhutdinov and Hinton in 2007. By training the algorithm to produce a low-dimensional binary code, all database entries could be stored in a hash table mapping binary code vectors to entries. This table would then support information retrieval by returning all entries with the same binary code as the query, or slightly less similar entries by flipping some bits from the query encoding.

Section 7

Communication Systems Autoencoders in communication systems are important because they help in encoding data into a more resilient representation for channel impairments, which is crucial for transmitting information while minimizing errors. In Addition, AE-based systems can optimize end-to-end communication performance. This approach can solve the several limitations of designing communication systems such as the inherent difficulty in accurately modeling the complex behavior of real-world channels.

Section 8

Image processing The characteristics of autoencoders are useful in image processing. One example can be found in lossy image compression, where autoencoders outperformed other approaches and proved competitive against JPEG 2000. Another useful application of autoencoders in image preprocessing is image denoising. Autoencoders found use in more demanding contexts such as medical imaging where they have been used for image denoising as well as super-resolution. In image-assisted diagnosis, experiments have applied autoencoders for breast cancer detection and for modelling the relation between the cognitive decline of Alzheimer's disease and the latent features of an autoencoder trained with MRI.